MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO, SWITCH ASSEMBLY

This specification sheet is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch housing described herein shall consist of this document and the latest issue of specification MIL-C-22442(AS).

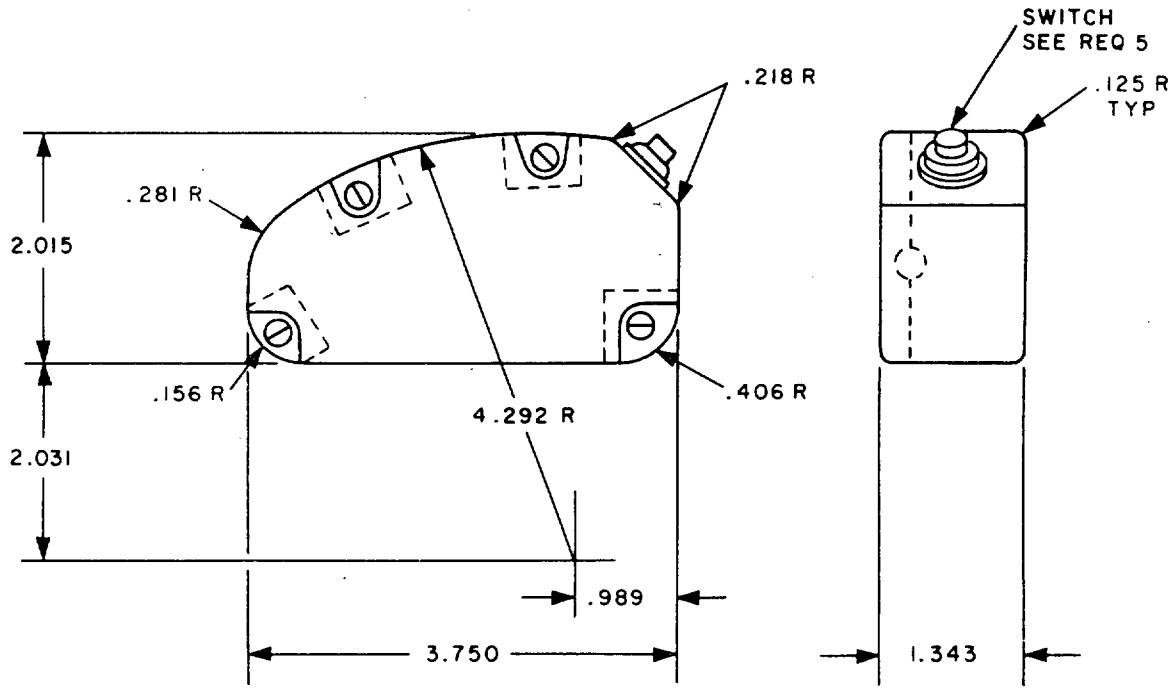


FIGURE I.

P/N M22442/1-1

FSC 5995

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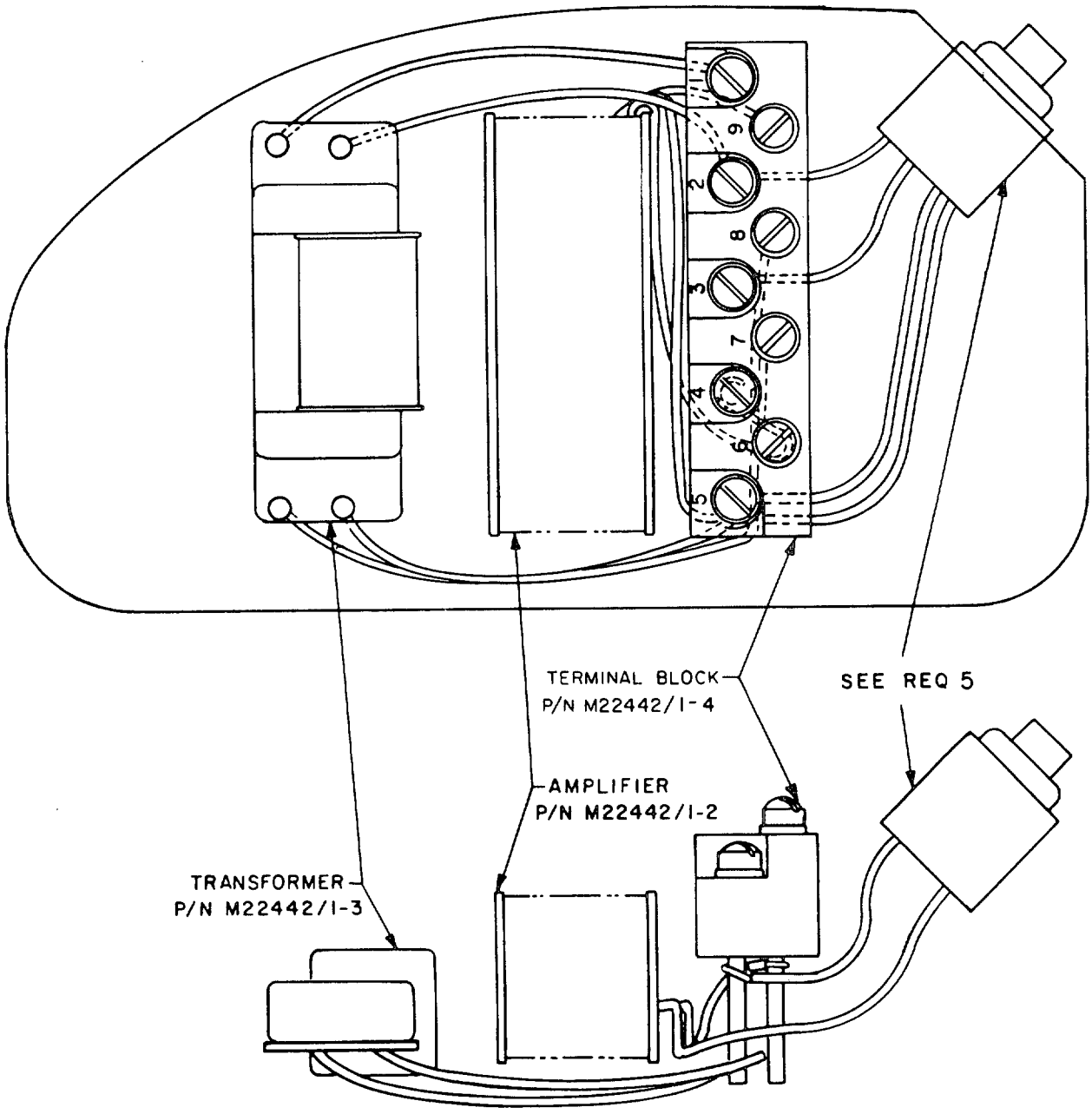


FIGURE II. SWITCH ASSEMBLY (TYPICAL)

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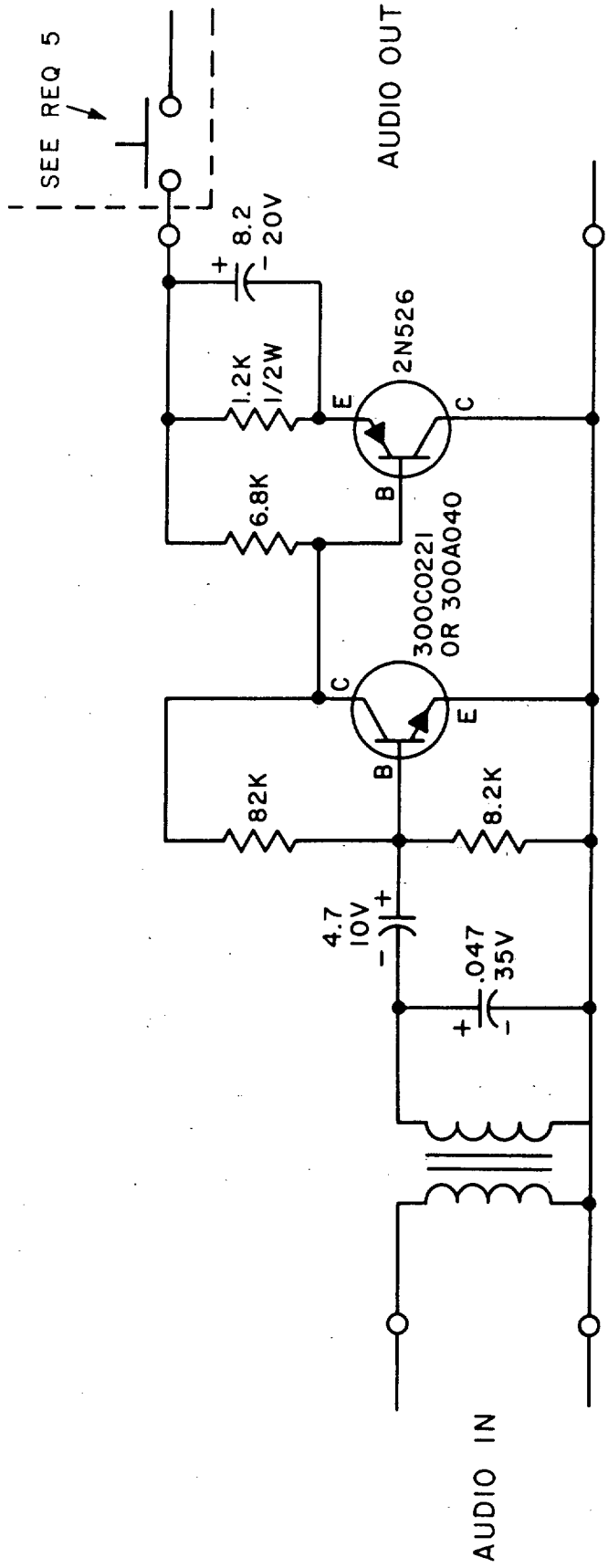


FIGURE III. SCHEMATIC AMPLIFIER AF (TYPICAL) P/N M22442/1-2

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APPLICABLE DOCUMENTS:

L-P-1183 Plastic Moulding Material, Acrylonitrile-Butadine-Styreen (ABS) Rigid

REQUIREMENTS:

1. The switch assembly shall contain a microphone amplifier, microphone keying switch, earphone matching transformer, and terminal block.

2. The switch housing shall be molded from molding plastic ABS rigid conforming to L-P-1183 Type II black or green.

3. The transformer shall have the following characteristics:

TRANSFORMER, AF M22442/1-3

a. POWER LOSS - Not greater than 2.5 dB at 1000 Hz.

b. FREQUENCY RESPONSE - The output response shall be within ±3 dB when referenced to the 1000 Hz level from 300 Hz to 7000 Hz. The input shall be adjusted for 13.4 volts.

(A) c. DISTORTION - Not greater than: 5% from 1000 to 6000 Hz.
15% from 300 to 1000 Hz.
The inputs shall be adjusted for 13.4 volts.

d. IMPEDANCE - Input 600 Ohm ± 15%
Output 7.5 Ohm ± 1 Ohm

DC RESISTANCE - Low Side 1.8 Ohm ± 0.2 Ohms
High Side 73.0 Ohm ± 3 Ohms

4. The amplifier assembly shall have the following characteristics:

AMPLIFIER, AF M22442/1-2

a. INPUT POWER - DC Voltage -- 27.5 ± 0.5
DC Current -- Not more than 30 milliamps

b. GAIN - 60 ± 3 dB when measured at 1000 Hz.
The input level shall be adjusted for 250 microvolts.

c. IMPEDANCE - Input 5 Ohm ± 1 Ohm
Output 100 Ohm ± 5%

d. DISTORTION - Less than 5%.

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e. FREQUENCY RESPONSE - The frequency response shall be within the limits shown below:

FREQUENCY	OUTPUT RELATIVE TO 1000 HZ	
Hz	Max. dB	Min. dB
300	-1.5	-6.0
500	+1.0	-3.0
750	+2.0	-2.0
1000	0	0
1500	+1.5	-1.5
2000	+1.5	-1.5
3000	+1.5	-1.5
4000	+1.5	-1.5
5000	+1.5	-2.0
6000	+1.5	-3.0

f. LINEARITY - The output voltage shall be a linear function of the input voltage within the input voltage levels of 5 microvolts to 500 microvolts.

5. Microphone Keying Switch. The microphone keying switch shall be a Micro-Switch (91929) type 2PB281 or equal.

6. The stay cord shall be firmly attached to the switch assembly.

7. The switch assembly shall not exceed 15% distortion.

NOTES:

1. Features shown in dotted lines are for reference only. Item need not be constructed in this manner.

2. Dimensions in inches. Unless otherwise specified, tolerances: linear $\pm$ .015.

Preparing Activity
Navy - AS
Project No. 5995-N110-1

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